

The Dobbertin (1999) Kasami paper, told in pictures

A visual map of the structural patterns
and how they propagate through the theorems

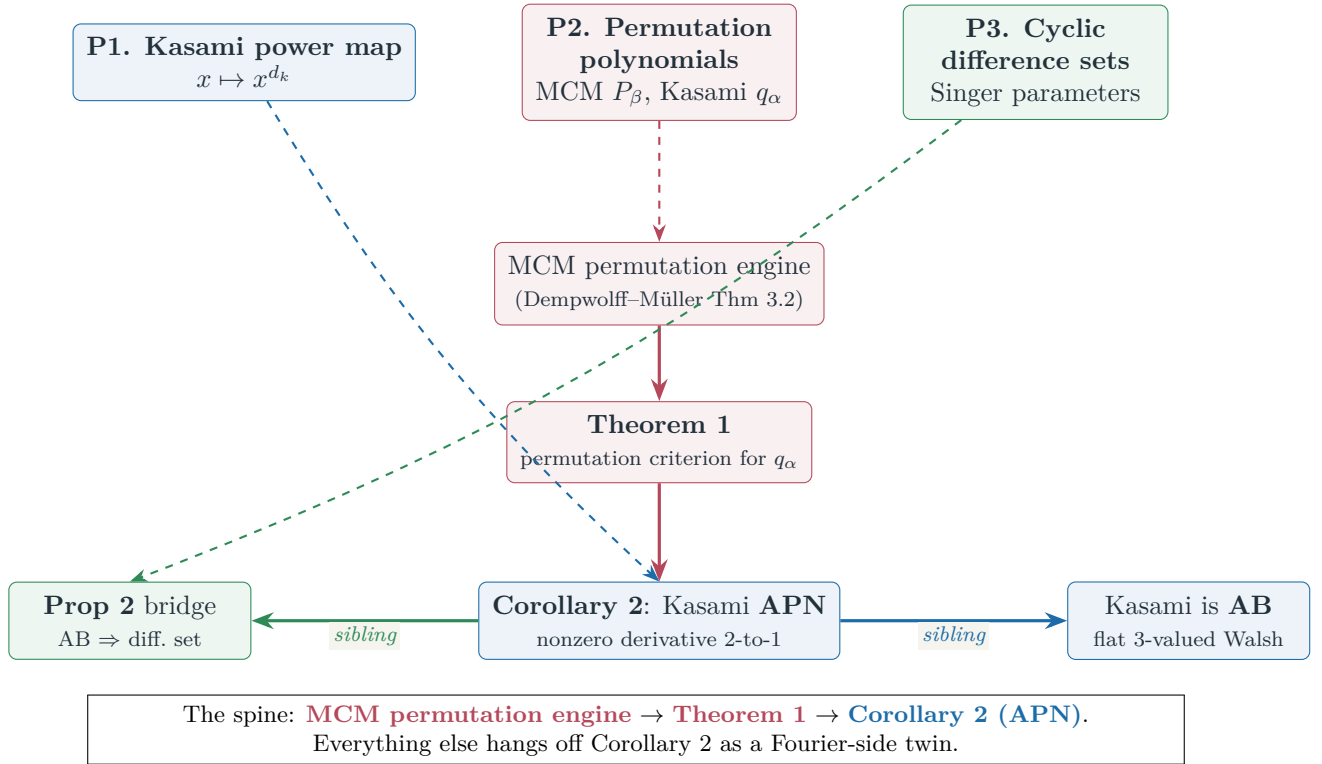
companion to RequestProject/ (every claim below is a **sorry-free** Lean fact)

*H. Dobbertin, “Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets”, NATO Sci. Ser. C
542, Kluwer, 1999.*

Field $L = \mathbb{F}_{2^n}$, Kasami exponent $d_k = 2^{2k} - 2^k + 1$, $\gcd(k, n) = 1$.

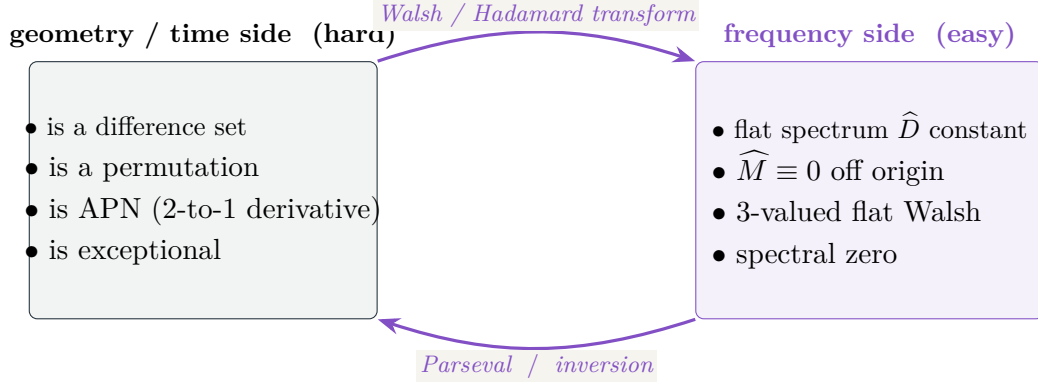
1. The paper in one picture: three pillars, one spine

Simple English: the title names three things. They are held up by a single chain of arrows. Follow the thick arrows.

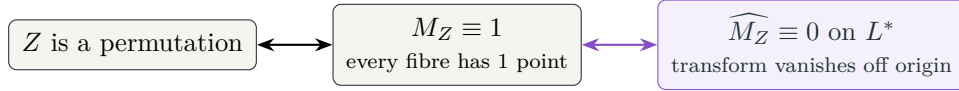


2. The one big idea: work on the Fourier side

Simple English: a hard property of a *map* becomes an easy property of its *Walsh transform*. “Hard, multiplicative” turns into “flat” or “zero”.



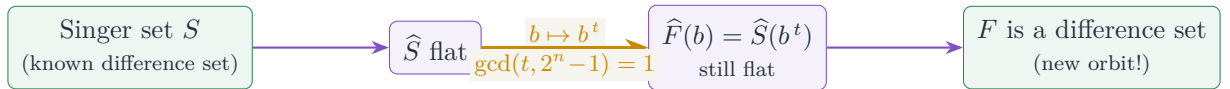
The multiplicity trichotomy (the cleanest instance):



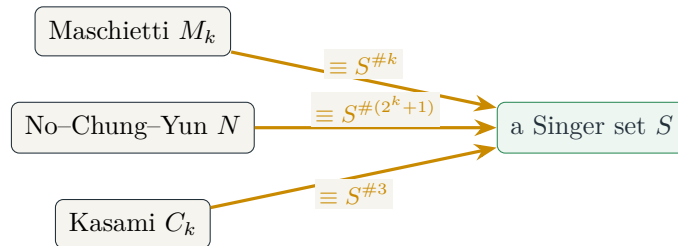
Lean echoes: `IsDifferenceSet.hasFlatSpectrum`, `KasamiAB.kasami_is_ab`, `APNLib.kasami_isAPN_diffUnif`.

3. Hadamard equivalence: slide the spectrum by a power

Simple English: if you already know one flat spectrum, you get a whole family for free — just relabel the frequency by a power t that permutes.



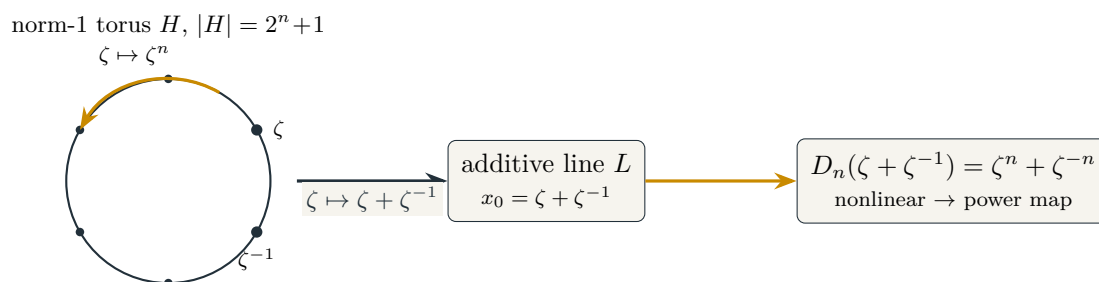
One lemma $\Rightarrow$ three theorems (paper eqns. (3)–(5)):



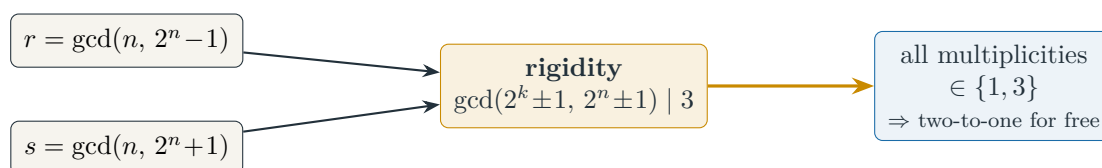
Hadamard-equivalence is *coarser* than design equivalence — that gap is exactly where the *new* difference sets live.

4. Linearize on a torus, then let arithmetic do the work

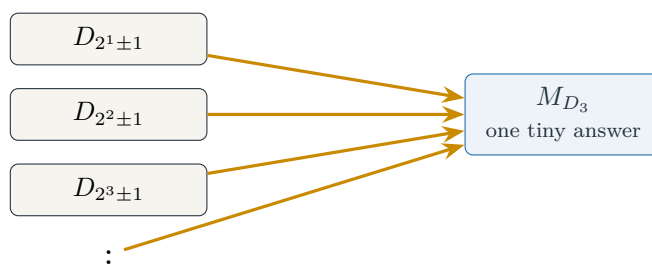
Simple English: the involution $\zeta \mapsto \zeta + \zeta^{-1}$ turns nonlinear Dickson maps into plain powers on a circle. Then two gcds decide everything.



Fibre sizes reduce to two gcds, and a rigidity forces them mod 3:

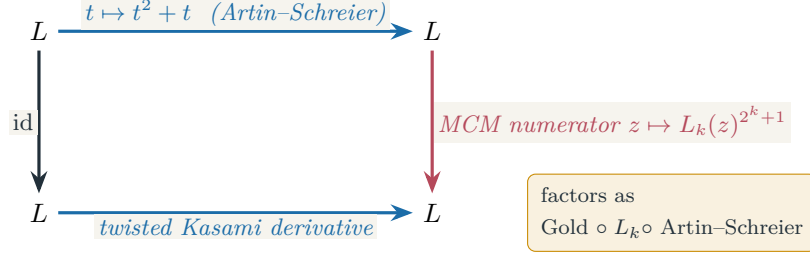


An entire infinite family collapses to the multiplicity of the tiny $D_3(X) = X^3 + X$: $M_{D_{2^k \pm 1}} = M_{D_3}$ (paper Prop 6 $\rightarrow$ Thm 7).



5. The MCM $\leftrightarrow$ Kasami connection = a square

Simple English: the Kasami derivative is literally the MCM numerator with an Artin–Schreier glue $t \mapsto t^2 + t$ in front. That is a commuting square.

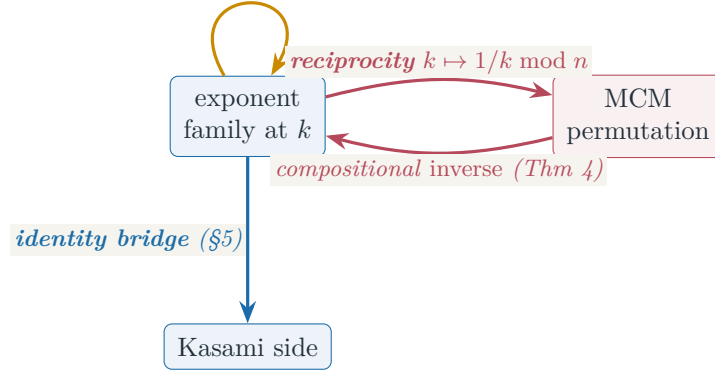


Key identity: $((t+1)^{d_k} + t^{d_k} + 1)(t^2 + t)^{2^k} = (t^{2^k} + t)^{2^k+1}$.
Lean: `MCMKasamiCat.mcmKasami_commSq, ... mcmNumerator_comp_artinSchreier.`

6. Three symmetries of the parameter k

Simple English: the exponent d_k can be moved three ways. Each move keeps the property you care about. They are different arrows in *one* equivalence groupoid.

$k \mapsto n - k, \quad d \mapsto 2^{2^k} \cdot d$
 additive-automorphism twist $\Rightarrow$ same differential spectrum

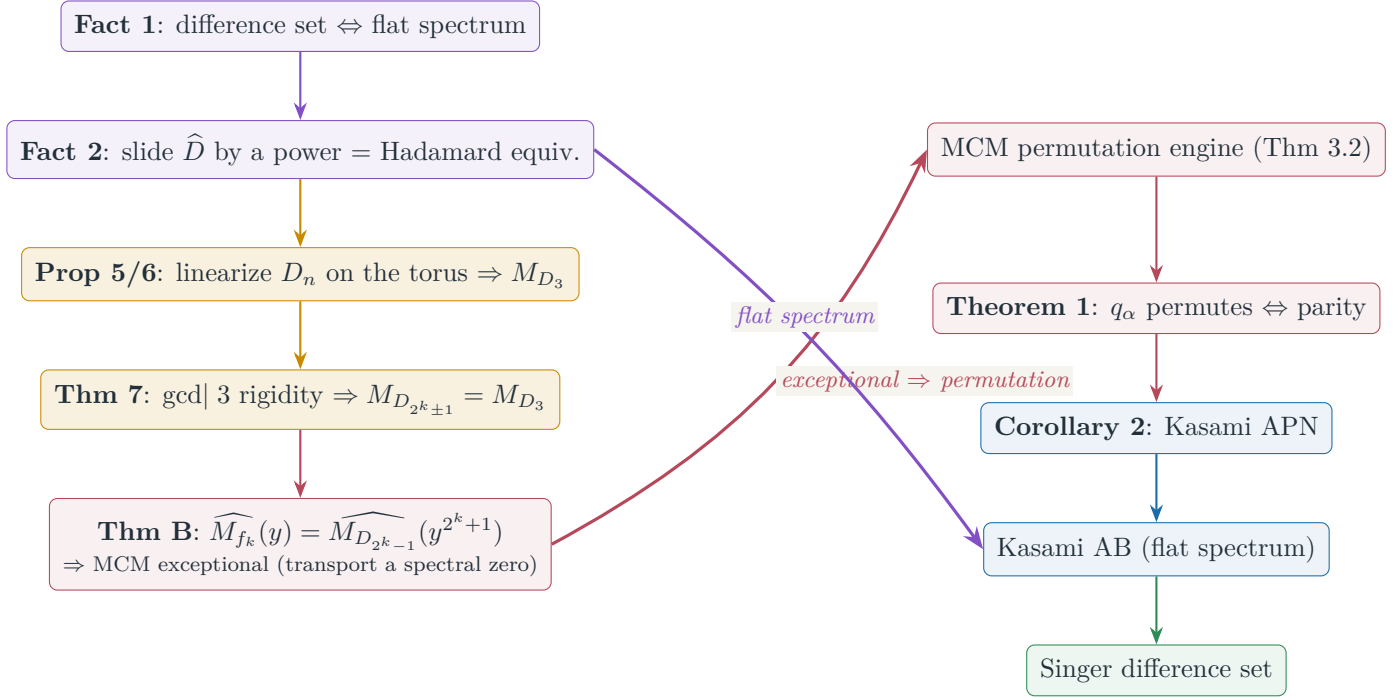


move	on k	on exponent	preserves
Frobenius / cyclotomic	$k \mapsto n - k$	$d \mapsto 2^{2^k} d$	differential spectrum (APN/AB)
MCM $\leftrightarrow$ Kasami (Thm 4)	$k \mapsto 1/k$	multiplicative inverse	permutation-ness
identity bridge (Cor 2)	—	Artin–Schreier glue	the 2-to-1 count

The Frobenius move is the paper’s one-line WLOG “replace k by $n - k$ if k is even; this does not change the APN property”. *Lean:* `FrobeniusGroupoid.kasamiComplementIso, ... diffUnifFunctor.`

7. How the patterns propagate through the paper

Simple English: read top to bottom. Each box uses the pattern above it. Colour = which structural idea is doing the work.



pattern	where	what it buys
Fourier duality	Facts 1–2, Thm B	geometry → transform identity
Hadamard equivalence	Fact 2, eqns (3)–(5)	one lemma → many difference sets; new orbits
linearize on a torus	Prop 5/6	fibre sizes = two gcds
gcd 3 rigidity	Thm 7	infinite family shares M_{D_3}
multiplicity trichotomy	before Thm B	uniform “is a permutation” test

One sentence. Every hard multiplicative property is dragged to the Fourier side, where it becomes a *flatness* or a *vanishing*, and is then carried between objects by *power-permutation reparametrization* — with Dickson maps linearized on a torus and a gcd | 3 rigidity forcing all the multiplicities into line.